

ajedrezUPV - Project Documentation

Artificial Intelligence Chess Engine Project

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Contents

1	Project Overview	2
2	Repository Structure	3
3	Chess Engine	3
4	Connect Four AI	4
5	Neural Networks	4
6	Technologies	4
7	Installation	4
8	License	4
9	Status	4
10	Contributing	4
11	Acknowledgments	4

1 Project Overview

ajedrezUPV is an AI chess project from UPV. Focus: decision algorithms, heuristic evaluation, game engine programming.

Key Features

- Custom chess engine (C++23)
- Alpha-Beta pruning algorithm
- UCI notation support
- Neural network integration
- Modular and extensible code
- Human vs AI gameplay

2 Repository Structure

Path	Description
.	Root
README.md	Main documentation
LICENSE	CC BY-NC 4.0
CMakeLists.txt	CMake build
Makefile	Make build
include/chess/	Public headers node_move.h evaluator.h general_evaluator.h opening_evaluator.h endgame_evaluator.h
src/engine/	Chess engine alphabeta_bot.cpp minimax_bot.cpp node_move.cpp/h evaluation/
src/neural/	Neural networks (C++) square_rook_network.cpp network_junior.hpp simd_network_wip.cpp
src/tools/	Data tools json_extractor.cpp json_extractor_junior.cpp
python/	Python tools model.py mnist_loader.py network_reference.py
data/	Data files dataset.csv
third_party/	Third-party headers json.hpp
docs/	Documentation FEN.pdf PROJECT_DOCUMENTATION.tex/.pdf
lichess_db/	Lichess DB chunk_000 to chunk_019.jsonl

3 Chess Engine

C++ implementation with Alpha-Beta algorithm using Disservin library.

Key Components

- `src/engine/alphabeta_bot.cpp` - Main game loop, UCI input, console display
- `node_move.cpp/h` - Game tree node with `alphaBeta()` method
- `general_evaluator.cpp` - Position evaluation
- `opening_evaluator.cpp` - Opening heuristics
- `endgame_evaluator.cpp` - Endgame heuristics

4 Connect Four AI

Neural network AI for Connect Four. Moved to its own repository: **connectUPV** (<https://github.com/LLPB-pixel/connectUPV>).

5 Neural Networks

- `neural_network/` - Main NN with jsonExtractor
- `neural_network_jr/` - Junior version

6 Technologies

C++: Disservin, nlohmann/json, STL. Python: TensorFlow, NumPy, scikit-learn, Jupyter.

7 Installation

CMake: `cmake -S . -B build && cmake --build build`

Make: `make setup && make all`

Run: `./bin/alphabeta_bot` (UCI notation: e2e4, etc.)

Connect Four is no longer part of this repository (see **connectUPV**).

8 License

CC BY-NC 4.0 - <https://creativecommons.org/licenses/by-nc/4.0/>

Allowed: Share, Adapt, Educational/Personal/Research use.

Not Allowed: Commercial use without permission.

9 Status

Completed: Chess engine, Alpha-Beta, evaluation system, victory check.

In Progress: Advanced NN for chess, move explorer, multi-threading, opening book.

10 Contributing

Llorenc Perez - perezllorenc@gmail.com - <https://github.com/LLPB-pixel/ajedrezUPV>

11 Acknowledgments

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<https://github.com/LLPB-pixel/ajedrezUPV>